VOLUME I

ANSWERS

UNIT-I

Choose the correct answer

1. b) $Al_2O_3.nH_2O$ 2. c) SO_2 3. c) $MgCO_3 \longrightarrow MgO + CO_2$
4. b) Al_2O_3 5. a) Al
6. d) Carbon and hydrogen are suitable reducing agents for metal sulphides.
7. c) A-iv , B-ii , C-iii , D-i
8. d)Electromagnetic separation
9. b) $Cu(s) + Zn^{2+}(aq) \longrightarrow Zn(s) + Cu^{2+}(aq)$
10. c) sodium
11. b) Infusible impurities to soluble impurities
12. c) Galena
13. a) Lower the melting point of alumina
14. a) Carbon reduction
15. c) Displacement with zinc
16. c) Mg
17. b) van Arkel process
18. d) Both (a) and (c)
19. d) In the metallurgy of gold, the metal is leached with dilute sodium chloride solution
20. b) Impure copper
21. b) ΔG^0 Vs T
22. c) $\left(\frac{\Delta G^0}{\Delta T}\right)$ is negative
23. b) $Al_2O_3 + 2Cr \longrightarrow Cr_2O_3 + 2Al$
24. b) The graph for the formation of CO_2 is a straight line almost parallel to free energy axis.

UNIT-2

Choose the correct answer:

1. c) basic

$$Na_2B_4O_7 + 7H_2O \rightleftharpoons \underset{\text{Strong base}}{2NaOH} + \underset{\text{Weak acid}}{4H_3BO_3}$$

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